

[Pivot & Unpivot Engine](#)

INSTANT

[Primary & Foreign Keys](#)[Data Analysis Diploma • Session 07](#)

EXCEL DATA MODELING & DASHBOARD DESIGN

Pivot/Unpivot · PK/FK Relationships · Group By · Star & Snowflake Schemas ·
UI/UX Rules

6 Goals**Core Outcomes**

Modeling • Schemas • UI

25 Slides**Curriculum Depth**

Data Modeling & UX

Visual Labs**Hands-on Practice**

PK/FK & Schemas Worksheets

[Start Session 07 →](#)[Star & Snowflake Schemas](#)[6 Rules of Dashboard Design](#)

— SESSION COMPETENCIES

What You Will Master Today

By the end of this session, you will master these core data modeling, transformation, and visual presentation skills:

01.



Pivot & Unpivot Operations

Convert unique values from rows to columns and transform wide datasets into normalized tall formats.

02.



Primary & Foreign Keys (PK/FK)

Establish unique record identifiers and relational links across enterprise database tables.

03.



Group By & Aggregations

Apply the Split–Apply–Combine engine to group data by keys and compute metric summaries.

04.



Append & Merge Queries

Combine datasets vertically across rows or horizontally using key–based joins.

05.



Star, Snowflake & Galaxy Schemas

Design central Fact tables connected to normalized or denormalized Dimension tables.

06.



UI/UX & Dashboard Rules

Apply 6 golden rules of colors, typography, layout flow, charts, background, and interactivity.



PART 1

Pivot & Unpivot Transformations

Reshaping Data Architecture for Aggregation and Analytical Readiness

RESHAPING ENGINE

Pivot vs. Unpivot Operations

Understanding how values transition between rows and columns during data restructuring:

Pivot: Rows to Columns

Unpivot: Wide to Long Format

PIVOT OPERATION DEFINITION

A data transformation operation that converts unique values from rows into separate columns, usually with aggregation (e.g., SUM, COUNT, AVG).

Raw Rows Format (Input)

Source

Attributes	Values
A1	V1
A2	V2
A3	V3
A1	V4
A2	V5
A3	V6

Pivoted Columns Format (Output)

Transformed

A1	A2	A3
V1	V2	V3
V4	V5	V6

DATA RESHAPING IN PRACTICE

Pivot & Unpivot Visual Workflows

Real-world data structures transitioning between wide presentation formats and long analytical formats:

Example 1: Yearly Credit Scores

Example 2: Monthly Products

Example 3: Country Profits

TRANSFORMING MULTI-YEAR HEADER COLUMNS INTO NORMALIZED ROWS

Unpivoted Wide Format

Year	CreditScore
2013	60
2015	50
2011	40
2012	30

Pivoted Matrix Format

2011	2012	2013	2015
40	30	60	50

— DATABASE INTEGRITY

Primary Key (PK) & Foreign Key (FK)

Building non-redundant relational links between parent and child tables in a database:

PRIMARY KEY (PK)

A column (or set of columns) that uniquely identifies each record in a table. It cannot contain duplicate or null values.

PARENT TABLE: USERS

PK: user_id

user_id (PK)	email	name
10	sadio@example.com	Sadio
11	mo@example.com	Mohamed
12	rinsola@example.com	Rinsola
13	amalie@example.com	Amalie

FOREIGN KEY (FK)

A column in one table that references the Primary Key of another table to create a relational connection.

CHILD TABLE: ORDERS

FK: user_id

order_no	user_id (FK)	product_sku
93	11	123
94	11	789
95	13	789
96	10	101



PART 2

Group By & Aggregation Mechanics

Summarizing Large Datasets using the Split-Apply-Combine Engine

DATA SUMMARIZATION

Group By Concept & Core Rules

Grouping raw observations by categorical keys to extract instant calculated metrics:

GROUP BY DEFINITION & CORE RULES

GroupBy is used to group data based on a specific column and perform calculations like sum, count, or average. It helps summarize data and extract meaningful insights quickly.

2 ESSENTIAL RULES

1- Columns to Group by (e.g. genre)

2- Aggregation Function (e.g. SUM of qty)

Raw Dataset → Grouped Total Summary

Raw Table (title, genre, qty)

book 1: adventure (4)
book 2: fantasy (5)
book 3: romance (2)
book 4: adventure (3)
book 5: fantasy (3)
book 6: romance (1)

Grouped Summary

adventure	7
fantasy	8
romance	3

— INTERNAL MECHANICS

How Group By Works Step-by-Step

The 3-stage computational flow executing under the hood during grouping:

1 SPLIT

2 APPLY

3 COMBINE

1. Input Dataset

key	data
A	1
B	2
C	3
A	4
B	5
C	6

2. Split & Apply (SUM)

Group A: [1, 4]

Sum = 5

Group B: [2, 5]

Sum = 7

Group C: [3, 6]

Sum = 9

3. Combine Result

key	total_sum
A	5
B	7
C	9



PART 3

Append & Merge Query Engine

Vertical Row Concatenation & Horizontal Key-Based Column Joins

— ROW STACKING

Append: Stacking Rows Vertically

Adding new records to the end of an existing dataset without altering original structures:

APPEND OPERATIONS (ROW CONCATENATION)

Adds rows from one dataset to the end of another dataset (Vertical Stacking). Retains original schema structure.

Input Datasets (Table 1 & Table 2)

Table A (Rows 1–3)

[A:1, B:2, C:3]

Table B (Rows 4–5)

[A:4, B:5, C:6]

Appended Output (Vertical Stack)

Combined Output Dataset

Row 1: [A:1, B:2, C:3]

Row 2: [A:4, B:5, C:6]

Row 3: [A:5, B:6, C:7]

— COLUMN JOINS

Merge: Joining Columns Horizontally

Enriching datasets by combining columns from multiple tables based on a shared key:

MERGE OPERATIONS (COLUMN JOINS)

Horizontal merge means adding columns from another dataset based on a shared key.

Input Datasets (Table 1 & Table 2)

Table A (Rows 1–3)

[A:1, B:2, C:3]

Table B (Rows 4–5)

[A:4, B:5, C:6]

Merged Output (Horizontal Columns)

Combined Output Dataset

Row 1: [A:1, B:2, C:3]

Row 2: [A:4, B:5, C:6]

Row 3: [A:5, B:6, C:7]



PART 4

Data Modeling & Architecture

Designing Data Storage, Normalization, Star Schemas & Snowflake Schemas

— FOUNDATIONAL ARCHITECTURE

Data Modeling Principles

Structuring entities, attributes, and relationships into a clear organizational blueprint:



Flat Table

CSV / Spreadsheets

- CSV & Text files
- Excel Spreadsheets
- Simple single-file data



Database

Relational (SQL)

- Relational (SQL)
- Non-Relational (NoSQL)
- Structured multi-table
- CRUD operations



Data Warehouse

Enterprise Analytics

- Optimized for analytics
- Historical data repository
- Schema-on-read/write
- OLAP Engines

— DATA ARCHITECTURE

Data Storing Classification

Categorizing data storage environments from flat files to enterprise data warehouses:



Flat Table

CSV / Spreadsheets

- CSV & Text files
- Excel Spreadsheets
- Simple single-file data



Database

Relational (SQL)

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Data Warehouse

Enterprise Analytics

- Optimized for analytics
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— WRITES VS READS

Normalized vs. Denormalized Data

Balancing write integrity and storage efficiency against read speed and reporting performance:

Normalized Data Structure

Multiple Small Tables (Write-Optimized)

Data organized into separate entities (Employee, Department, Location) connected via Foreign Keys to minimize duplication.

- Reduces Redundancy
- More Joins Needed (Slower Reads)
- Excellent for Writes & Data Integrity

Denormalized Data Structure

Single Combined Table (Read-Optimized)

Data combined into fewer tables (with controlled duplication) to eliminate JOIN overhead and speed up reporting queries.

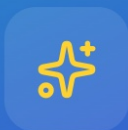
- Single Combined Wide Table
- Increases Controlled Redundancy
- Fewer Joins (Faster Reads & Analytics)

Trade-off: Writes & Integrity ◀=====▶ Reads & Reporting Speed

— MODEL COMPARISON

Star Schema vs. Snowflake Schema

The two foundational dimensional modeling designs for analytics and BI platforms:



STAR SCHEMA

De-normalized dimension tables connected directly to one central Fact Table.



SNOWFLAKE SCHEMA

Normalized dimension tables decomposed into secondary sub-dimension hierarchies.

— CENTRAL FACT PATTERN

Star Schema Architecture

Connecting a central Fact table to de-normalized Dimension tables for high-speed queries:

Star Schema Definition: A data warehouse structure where one central fact table is connected to multiple dimension tables to make data analysis simple and fast.



— MULTI-LEVEL HIERARCHIES

Snowflake Schema Architecture

Normalizing Dimension tables into secondary sub-dimensions to eliminate redundant attributes:

Snowflake Schema Definition: Dimension tables are normalized into multiple related tables, making the data more organized but more complex to query compared to a star schema.



Star Schema vs. Snowflake Comparison

Evaluating key trade-offs across simplicity, Power BI speed, redundancy, and scalability:

Aspect	Star Schema	Snowflake Schema
Simplicity	Simpler	More complex
Performance in Power BI	Higher performance	Lower performance
Redundancy	High redundancy	Low redundancy
Design	De-normalized	Normalized
Common Usage	Small to medium datasets	Large enterprise data warehouses

— FACT CONSTELLATION PATTERN

Galaxy Schema (Fact Constellation)

Linking multiple central Fact tables to shared conformable Dimension tables:

Galaxy Schema Definition: Multiple fact tables share dimension tables, used to model complex enterprise business processes.

SALES Fact Table

INVENTORY Fact Table

Shared Dimension Tables (Product, Time, Location, Supplier)



PART 5

UI/UX & Dashboard Design Principles

Crafting High-Impact, Readable & Interactive Visual Reports

DESIGNER ESSENTIALS

Websites & Tools Every Designer Needs

Essential web resources for icons, color palettes, object removal, and visual inspiration:

FOR ICONS

[Flaticon](#)
[Icons8](#)

FOR DESIGN & COLORS

[Figma](#)
[Adobe Color](#)
[Remove Background](#)
[Remove Objects](#)
[Colourco](#)

VISUAL FEEDING & INSPIRATION

[Dribbble](#)
[NovyPro](#)

— VISUAL DESIGN PILLARS

6 Golden Rules of Dashboard Design

Core principles for visual clarity, typographic hierarchy, layout flow, charts, and interactivity:

01.



01. Colors

Use 3–4 main colors only. Prefer shades of the same color and maintain strict consistency across all pages.

02.



02. Typography (Texts)

Choose a clear font. Hierarchy: Titles large & bold, KPIs medium, Details smaller. Keep labels short & meaningful.

03.



03. Background

Use a neutral background. The background should always highlight the data, never compete with it.

04.



04. Layout & Structure

Organize visuals by story flow (F-shape or Z-shape). Keep equal spacing and avoid overcrowding.

05.



05. Charts

Pick the right chart for the story. Avoid 3D charts, add clear titles, use data labels, and highlight key numbers.

06.



06. Interactivity

Place slicers & filters in one clean area. Use navigation buttons, tooltips on hover, and enable drill-down/through.

INSTANT

✔ SESSION 07 COMPLETED SUCCESSFULLY

THANK YOU

Thank you for your active participation!

Next Session: Python Fundamentals – Syntax, Variables & Data Types

↺ Restart Presentation

Open Session 09: Python Fundamentals →